Weibull Model Project

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1 Weibull Model

1.1 Weibull Density

$f(x;\xi,\theta) = \frac{\xi}{\theta}(\frac{x}{\theta})^{\xi-1}e^{-(\frac{x}{\theta})^{\xi}}\mathbf{1}_{x\geq 0}$

1.2 Likelihood

$$l(x_{(1)},...,x_{(n)}) = [\prod_{l=1}^{k} f(x_{(i)})][S(x_{(k)})]^{n-k} = \prod_{i=1}^{k}(\frac{\xi}{\theta}(\frac{x_{(i)}}{\theta})^{\xi-1}e^{-(\frac{x_{(i)}}{\theta})^{\xi}}\mathbf{1}_{x_{(i)}\geq 0})(\int_{x_{(k)}}^{+\infty} f(x;\xi,\theta)dx)^{n-k}$$

$$(\int_{x_{(k)}}^{+\infty} f(x;\xi,\theta)dx)^{n-k} = (\int_{x_{(k)}}^{+\infty} \frac{\xi}{\theta}(\frac{x}{\theta})^{\xi-1}e^{-(\frac{x}{\theta})^{\xi}}dx)^{n-k} = ([-e^{-(\frac{x}{\theta})^{\xi}}]_{x_{(k)}}^{+\infty})^{(n-k)} = e^{-(n-k)(\frac{x_{(k)}}{\theta})^{\xi}}$$

$$l(x_{(1)},...,x_{(n)}) = \mathbf{1}_{\{x_{(1)}\geq 0\}}\frac{\xi^k}{\theta^{k\xi}}(\prod_{i=1}^{k} x_{(i)}^{\xi-1})e^{-\frac{1}{\theta^{\xi}}T(\xi)}$$

$T(\xi) = \sum_{i=1}^{k} x_{(i)}^{\xi} + (n-k)x_{(k)}^{\xi}$

1.3 A priori distribution on θ

$\theta \sim U_{[0,\delta]}$

1.3.1 Computation of a posteriori law

<u>Computation details (n=1)</u>:

$$\pi(\theta|x) = \frac{f(x|\theta)\pi(\theta)}{\int_0^{\delta} f(x|\theta)\pi(\theta)d\theta} = \frac{\frac{\xi}{\theta^{\xi}}x^{\xi-1}e^{-(\frac{x}{\theta})^{\xi}}\mathbf{1}_{\mathbf{0}\leq\theta\leq\delta}}{\int_0^{\delta}\frac{\xi}{\theta^{\xi}}x^{\xi-1}e^{-(\frac{x}{\theta})^{\xi}}d\theta}$$

$$\pi(\theta|x) = \frac{\frac{\xi}{\theta^{\xi}}x^{\xi-1}e^{-(\frac{x}{\theta})^{\xi}}\mathbf{1}_{\mathbf{0}\leq\theta\leq\delta}}{\int_0^{\delta}\frac{\xi}{\theta^{\xi}}x^{\xi-1}e^{-(\frac{x}{\theta})^{\xi}}d\theta} = \frac{\frac{\xi}{\theta^{\xi}}x^{\xi-1}e^{-(\frac{x}{\theta})^{\xi}}\mathbf{1}_{\mathbf{0}\leq\theta\leq\delta}}{\int_0^{+\infty}\frac{\xi}{\theta^{\xi}}x^{\xi-1}e^{-(\frac{x}{\theta})^{\xi}}d\theta - \int_{\delta}^{+\infty}\frac{\xi}{\theta^{\xi}}x^{\xi-1}e^{-(\frac{x}{\theta})^{\xi}}d\theta}$$

$$\left[\begin{array}{c} u = (\frac{x}{\theta})^{\xi} \\ |\frac{du}{d\theta}| = \frac{\xi x^{\xi}}{\theta^{\xi+1}} \\ \theta = xu^{-\frac{1}{\xi}} \\ (\frac{x}{\theta})^{\xi} \in [0,(\frac{x}{\delta})^{\xi}] \end{array}\right.$$

$$\pi(\theta|x) = \frac{\frac{\xi}{\theta^\xi} x^{\xi-1} e^{-(\frac{x}{\theta})^\xi} \mathbf{1}_{\mathbf{0} \leq \theta \leq \delta}}{\int_0^{+\infty} e^{-u} u^{(1-\frac{1}{\xi})-1} du - \int_{(\frac{x}{\delta})^\xi}^{+\infty} e^{-u} u^{(1-\frac{1}{\xi})-1} du}$$

$$\pi(\theta|x) = \frac{\xi x^{\xi-1} e^{-(\frac{x}{\theta})^\xi} \mathbf{1}_{\mathbf{0} \leq \theta \leq \delta}}{\theta^\xi (\Gamma(1-\frac{1}{\xi}) - \Gamma(\frac{x^\xi}{\delta^\xi}; 1-\frac{1}{\xi}, 1))}$$

<u>Computation details (n>1)</u>:

$$\pi(\theta|x) = \frac{\xi e^{-\frac{T(\xi)}{\theta^\xi}} \mathbf{1}_{(\theta) \in [\mathbf{0}, \delta]}}{\theta^{k\xi} T(\xi)^{\alpha(\xi)} (\Gamma(k-\frac{1}{\xi} - \Gamma(\frac{T(\xi)}{\delta^\xi}; k-\frac{1}{\xi}, 1))}$$

1.3.2 Computation of the Bayes Estimator under Quadratic Cost Assumption

$\mathbf{E}^{\pi(.|x)}[\theta] = \int_0^\delta \theta \pi(\theta|x) d\theta$

$\mathbf{E}^{\pi(.|x)}[\theta] = \frac{\int_0^\delta \theta^{-\xi+1} \xi x^{\xi-1} e^{-(\frac{x}{\theta})^\xi} d\theta}{(\Gamma(1-\frac{1}{\xi}) - \Gamma((\frac{x}{\delta})^\xi; 1-\frac{1}{\xi}, 1)}$

$$\left[\begin{array}{c} u = (\frac{x}{\theta})^\xi \\ |\frac{du}{d\theta}| = \frac{\xi x^\xi}{\theta^{\xi+1}} \\ \theta^2 = x^2 u^{-\frac{2}{\xi}} \\ (\frac{x}{\theta})^\xi \in [0, (\frac{x}{\delta})^\xi] \end{array} \right.$$

$\mathbf{E}^{\pi(.|x)}[\theta] = \frac{(\Gamma(k-\frac{2}{\xi}) - \Gamma(\frac{T(\xi)}{\delta^\xi}; k-\frac{2}{\xi}, 1))}{\phi(\xi) T(\xi)^{k-\frac{2}{\xi}}}$

$$\left[\begin{array}{c} \phi(\xi) = (T(\xi))^{\alpha(\xi)} (\Gamma(k-\frac{1}{\xi} - \Gamma(\frac{T(\xi)}{\delta^\xi}; k-\frac{1}{\xi}, 1)) \\ \alpha(\xi) = \frac{1}{\xi} - (k-1) \end{array} \right.$$

1.3.3 Variance of θ under a posteriori distribution and confidence interval

Variance $V(\theta|x) = \frac{(\Gamma(k-\frac{3}{\xi}) - \Gamma(\frac{T(\xi)}{\delta^\xi}; k-\frac{3}{\xi}, 1)}{\phi(\xi) T(\xi)^{k-\frac{3}{\xi}}} - (\frac{(\Gamma(k-\frac{2}{\xi}) - \Gamma(\frac{T(\xi)}{\delta^\xi}; k-\frac{2}{\xi}, 1))}{\phi(\xi) T(\xi)^{k-\frac{2}{\xi}}})^2$

Confidence Interval We propose two solutions:

<u>Solution 1:</u> $\mathbf{E}[\theta|\mathbf{X}] \pm z_{1-\frac{\alpha}{2}} \sqrt{V(\theta|X)}$

<u>Solution 2:</u> $\pi(\theta \in [q_1, q_2]|X) = 1 - \alpha \Leftrightarrow$

$\Gamma(\phi_1(\xi); k-\frac{1}{\xi}, 1) - \Gamma(\phi_2(\xi); k-\frac{1}{\xi}, 1) = (1-\alpha)\phi(\xi) T(\xi)^{k-\frac{1}{\xi}}$

$$\left[\begin{array}{c} \phi_1(\xi) = \frac{T(\xi)}{q_2^\xi} \\ \phi_2(\xi) = \frac{T(\xi)}{q_1^\xi} \end{array} \right.$$

We can use Monte Carlo methods to compute integrals.

1.4 Inverse gamma a priori distribution

$\pi(\theta) = \mathbf{1}_{R^+}(\theta)(\frac{\mu}{\theta})^{\nu+1}\frac{e^{-\frac{\mu}{\theta}}}{\mu\Gamma(\nu)}$

1.4.1 Computation of the joint a posteriori law

$\pi(\theta,\xi|x) =^{\theta\perp\xi} \pi(\theta|x)\pi(\xi|x)$

$\pi(\theta|x) = \frac{\frac{e^{-(\frac{x}{\theta})^{\xi}-\frac{\mu}{\theta}}}{\theta^{\xi+\nu+1}}\mathbf{1}_{\theta\geq 0}}{\int_0^{+\infty}\frac{e^{-(\frac{x}{\theta})^{\xi}-\frac{\mu}{\theta}}}{\theta^{\xi+\nu+1}}d\theta}$

$$\pi(\theta|x) = \frac{\frac{e^{-(\frac{x}{\theta})^{\xi}-\frac{\mu}{\theta}}}{\theta^{\xi+\nu+1}}\mathbf{1}_{\theta\geq 0}}{\int_0^{+\infty}\int_0^{\pi/2} r^{2\nu+3-\frac{1}{\xi}}e^{-x^{\xi}(r[(cos(\theta)+sin(\theta))+\mu(cos(\theta)^{\frac{1}{\xi}}+sin(\theta)^{\frac{1}{\xi}})])}(\frac{sin(2\theta)}{2})^{\nu}(sin^2(\theta)-cos^2(\theta))d\theta dr}$$

$$\pi(\xi|x) = \frac{\frac{\xi}{\theta}(\frac{x}{\theta})^{\xi-1}e^{-(\frac{x}{\theta})^{\xi}}\frac{1}{\beta-\alpha}\mathbf{1}^{(\xi)}_{[\alpha,\beta]}}{\int_\alpha^\beta \frac{\xi}{\theta}(\frac{x}{\theta})^{\xi-1}e^{-(\frac{x}{\theta})^{\xi}}\frac{1}{\beta-\alpha}d\xi}$$

$$\left[\begin{array}{c} u = (\frac{x}{\theta})^{\xi} \\ |\frac{du}{d\xi}| = log(\frac{x}{\theta})(\frac{x}{\theta})^{\xi} \\ \xi = \frac{log(u)}{log(\frac{x}{\theta})} \\ u \in [(\frac{x}{\theta})^{\alpha}\wedge(\frac{x}{\theta})^{\beta}, (\frac{x}{\theta})^{\beta}\vee(\frac{x}{\theta})^{\xi}] \end{array}\right.$$

$$\pi(\xi|x) = \frac{\xi x^{\xi-1}e^{-(\frac{x}{\theta})^{\xi}}\mathbf{1}^{(\xi)}_{[\alpha,\beta]}}{\frac{1}{(log(\frac{x}{\theta}))^2}\int_{(\frac{x}{\theta})^{\alpha}\wedge(\frac{x}{\theta})^{\beta}}^{(\frac{x}{\theta})^{\beta}\vee(\frac{x}{\theta})^{\alpha}} log(u)e^{-u}du}$$

$M = (\frac{x}{\theta})^{\beta}\vee(\frac{x}{\theta})^{\alpha}$
$m = (\frac{x}{\theta})^{\beta}\wedge(\frac{x}{\theta})^{\alpha}$

$$\pi(\xi|x) = \frac{\xi log(\frac{x}{\theta})^2 x^{\xi-1}e^{-(\frac{x}{\theta})^{\xi}}\mathbf{1}^{(\xi)}_{[\alpha,\beta]}}{((log(m)e^{-m}-log(M)e^{-M})+(Ei(M)-Ei(m))}$$

1.4.2 Computation of the Bayes Estimator under Quadratic Cost Assumption

$\mathbf{E}^{\pi((\theta,\xi)|\mathbf{x})}[(\theta,\xi)] = (\mathbf{E}^{\pi((\theta,\xi)|\mathbf{x})}[\theta], \mathbf{E}^{\pi((\theta,\xi)|\mathbf{x})}[\xi]) =^{\theta\perp\xi} (\mathbf{E}^{\pi(.|\mathbf{x})}[\theta], \mathbf{E}^{\pi(.|\mathbf{x})}[\xi])$

$\mathbf{E}^{\pi(.|\mathbf{x})}[\theta] = \frac{1}{\int_0^{+\infty}\frac{e^{-(\frac{x}{\theta})^{\xi}-\frac{\mu}{\theta}}}{\theta^{\xi+\nu+1}}d\theta}\int_0^{+\infty}\frac{\theta e^{-(\frac{x}{\theta})^{\xi}-\frac{\mu}{\theta}}}{\theta^{\xi+\nu+1}}d\theta$

$\mathbf{E}^{\pi(.|\mathbf{x})}[\xi] = \frac{\int_\alpha^\beta \xi^2 log(\frac{x}{m})^2 x^{\xi-1}e^{-(\frac{x}{\theta})^{\xi}}d\xi}{((log(m)e^{-m}-log(M)e^{-M})+(Ei(M)-Ei(m))} = \frac{log(\frac{x}{m})^2\int_\alpha^\beta \xi^2 x^{\xi-1}e^{-(\frac{x}{\theta})^{\xi}}d\xi}{((log(m)e^{-m}-log(M)e^{-M})+(Ei(M)-Ei(m))}$

$\Leftrightarrow$

$\mathbf{E}^{\pi(.|\mathbf{x})}[\xi] = \frac{log(\frac{x}{m})^2\int_\alpha^\beta \xi^2 e^{(\xi-1)log(x)-(\frac{x}{\theta})^{\xi}}d\xi}{((log(m)e^{-m}-log(M)e^{-M})+(Ei(M)-Ei(m))}$

1.5 Non informative distribution under Jeffrey's Rule

1.5.1 Log-Likelihood function

$L(x_{(1)},...,x_{(n)};\theta,\xi) = log((\prod_{i=1}^{k} f(x_{(i)};\theta,\xi))[P(X > x_{(k)}]^{n-k})$

$= \sum_{i=1}^{k} log(f(x_{(i)};\theta,\xi)) + (n-k)log(P(X > x_{(k)}))$

$= \sum_{i=1}^{k} log(\frac{\xi}{\theta}(\frac{x_{(i)}}{\theta})^{\xi-1}e^{-(\frac{x_{(i)}}{\theta})^{\xi}}) + (n-k)log(\int_{x_{(k)}}^{+\infty} \frac{\xi}{\theta}(\frac{x}{\theta})^{\xi-1}e^{-(\frac{x}{\theta})^{\xi}}dx)$

$= (\xi-1)\sum_{i=1}^{k} log(x_{(i)}) - k(\xi-1)log(\theta) + klog(\xi) - [(\sum_{i=1}^{k}(\frac{x_{(i)}}{\theta})^{\xi} + (n-k)(\frac{x_{(k)}}{\theta})^{\xi}]$

1.5.2 Score function

$\mathrm{U}(\theta,\xi) = (\frac{\partial L}{\partial\theta},\frac{\partial L}{\partial\xi})$

$\frac{\partial L}{\partial\theta} = -\frac{k(\xi-1)}{\theta} + \frac{\xi}{\theta^{\xi+1}}(\sum_{i=1}^{k} x_{(i)}^{\xi} + (n-k)x_{(k)}^{\xi})$

$\frac{\partial L}{\partial\xi} = \sum_{i=1}^{k} log(x_{(i)}) - klog(\theta) + \frac{k}{\xi} - (\sum_{i=1}^{k} log(\frac{x_{(i)}}{\theta})(\frac{x_{(i)}}{\theta})^{\xi} + (n-k)log(\frac{x_{(k)}}{\theta})(\frac{x_{(k)}}{\theta})^{\xi})$

1.5.3 Fisher Information Matrix

$I(\theta,\xi) = \mathbf{E}[-\frac{\partial^2 L}{\partial\theta^2}]$

$\frac{\partial^2 L}{\partial\theta^2} = \frac{(\xi-1)k}{\theta^2} - \frac{\xi(\xi+1)}{\theta^{\xi+2}}T(\xi)$

$\frac{\partial^2 L}{\partial\xi^2} = -\frac{k}{\xi^2} - \sum_{i=1}^{k}(log((\frac{x_{(i)}}{\theta})^{\xi})^2(\frac{x_{(i)}}{\theta})^{\xi}) + (n-k)log(\frac{x_{(k)}}{\theta})^2(\frac{x_{(k)}}{\theta})^{\xi}$

$\frac{\partial^2 L}{\partial\xi\partial\theta} = -\frac{k}{\theta} + \sum_{i=1}^{k}(\frac{x_{(i)}^{\xi}}{\theta^{\xi+1}})(1+\xi log(\frac{x_{(i)}}{\theta})) + \frac{(n-k)(x_{(k)}^{\xi})}{\theta^{\xi+1}}(1+\xi log(\frac{x_{(k)}}{\theta})$

1.5.4 Jeffrey's Rule TO UPDATE

$\pi(\theta,\xi) \propto \sqrt{|I((\theta,\xi))|}$

$$I((\theta,\xi))_1 = \begin{bmatrix} -\frac{\xi k}{\theta^2} + \frac{\xi(\xi+1)}{\theta^{\xi+2}}(\sum_{i=1}^{k}(x_{(i)}^{\xi} + (n-k)(x_{(k)})^{\xi}) \\ \frac{k}{\theta} - \frac{1}{\theta^{\xi+1}}(\sum_{i=1}^{k}[x_{(i)}(1+\xi log(\frac{x_{(i)}}{\theta})] - (n-k)(x_{(k)})^{\xi}(1+\xi log(\frac{x_{(k)}}{\theta})) \end{bmatrix}$$

$$I((\theta,\xi))_2 = \begin{bmatrix} \frac{k}{\theta} - \frac{1}{\theta^{\xi+1}}(\sum_{i=1}^{k}[x_{(i)}(1+\xi log(\frac{x_{(i)}}{\theta})] + (n-k)(x_{(k)})^{\xi}(1+\xi log(\frac{x_{(k)}}{\theta})) \\ \frac{k}{\xi^2} + \sum_{i=1}^{k}(log^2((\frac{x_{(i)}}{\theta})^{\xi})(\frac{x_{(i)}}{\theta})^{\xi}) + (n-k)log^2(\frac{x_{(k)}}{\theta})(\frac{x_{(k)}}{\theta})^{\xi} \end{bmatrix}$$

References